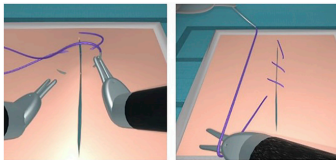

Mathematical Modeling and Motion Planning for Robotic Manipulation of Deformable Linear Objects

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Specialty 1.2.2	Mathematical modeling, numerical methods and software packages

Thesis for the degree of Candidate of Technical Sciences



Wire-harness assembly



Surgical-thread manipulation

A **deformable linear object (DLO)** is a cable, wire, rope, hose, or thread whose complete shape changes during manipulation.

- Distributed compliance makes the cable shape high-dimensional; the two endpoints do not determine it.
- Material and dynamic parameters are difficult to identify for each object.
- Local shape controllers may fail for large transformations, including a reversal of concavity.
- Collision-free endpoint paths or static cable shapes do not guarantee collision-free motion of every cable segment.

Research problem. Control and route the complete planar DLO with two robots using its geometry, without prior identification of stiffness or damping.

Source: Dissertation, Introduction and Chapter 1.

Goal. Develop mathematical models and effective numerical methods for dual-arm shape control and routing of DLOs under large planar deformations and workspace obstacles, without prior identification of dynamic or material parameters.

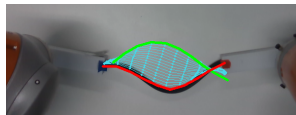
Research objectives

1. Reconstruct and track a marker-free planar cable from images.
2. Generate length-feasible intermediary configurations, including opposite-concavity transformations.
3. Model local robot–cable kinematics and calculate bounded dual-arm motion.
4. Plan the complete cable route among static obstacles with valid swept motion and endpoint correspondence.
5. Implement and verify the developed models and numerical methods.

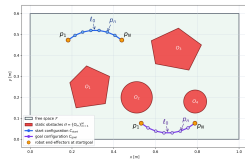
Object: planar dual-arm robotic manipulation of a DLO.

Subject: mathematical models, numerical methods, and algorithms for representation, simulation, perception, deformation planning, control, and routing.

Source: Dissertation, Introduction: goal, object, subject, and objectives.



Large shape deformation



Routing among obstacles

The cable centerline is represented by ordered points $\mathbf{p} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^T \in \mathbb{R}^{2N}$, with nominal segment length l_s .

Problem	Given	Required	Validity conditions
Shape control	Current shape $\mathbf{p}$, target $\mathbf{t}$, and two end-effector poses $\mathbf{r}$.	Reference profiles $\mathbf{p}^0, \dots, \mathbf{p}^K = \mathbf{t}$ and bounded end-effector velocities $\dot{\mathbf{r}}$.	Approximately fixed discrete length, bounded robot velocities, and minimization of aggregate shape error.
Global routing	Ordered start and goal configurations C_s, C_g and known static obstacles $\mathcal{O}$.	A sequence $\mathcal{P} = (C_0, \dots, C_K)$, from start to goal.	Every configuration is statically admissible; every neighboring transition is collision-free and preserves endpoint correspondence.

Validation boundary. Chapter 3 uses image feedback in physical shape-control experiments. Chapter 4 starts from prescribed ordered configurations and known obstacle geometry and is validated computationally and in MuJoCo.

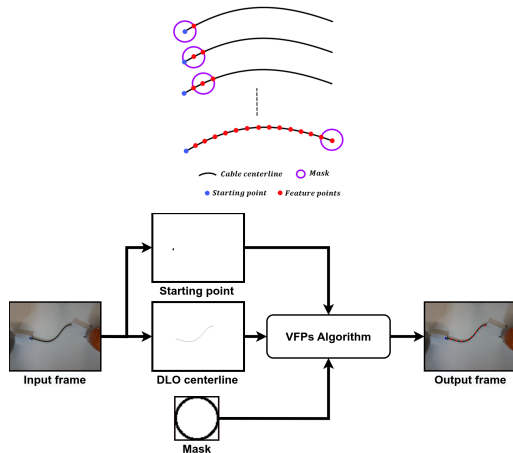
1. **Virtual Feature Points (VFPs):** an ordered, approximately equidistant point representation reconstructed from an image of a marker-free cable.
2. **Intermediary-shape planning:** The Angular and Linear Interpolation-based Planner (ALI) for large deformation of shape control.
3. **Shape control:** a diminishing-rigidity Jacobian and constrained optimization for bounded two-arm motion without identified stiffness and bending parameters of the cable.
4. **Global Cable Routing (GCR):** an ordered configuration-space model and numerical method with guided sampling, Artificial Potential Field (APF) bias, projected relaxation, and swept-transition validation.
5. **Software and verification:** Vision-based cable detection and tracking, MATLAB simulations, a C++ GCR implementation, physical KUKA experiments, and MuJoCo validation.

Source: Dissertation, Introduction: scientific novelty and main results.

1. Detect the marked tool and identify the starting cable point.
2. Segment the marker-free cable using edges and morphology.
3. Apply Guo–Hall thinning to extract a one-pixel centerline.
4. Place the center of a circular mask on the starting point.
5. Record the forward intersection of the mask and centerline as a new feature point.
6. Slide the mask to the new feature point.
7. Repeat Steps 5 and 6 until the cable end is reached.

Result. VFPs reconstruct an ordered, approximately equidistant cable state from one top-view camera image.

$$\mathbf{p} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^T, \quad \|\mathbf{p}_{n+1} - \mathbf{p}_n\| \approx l_s.$$



Aim: Decompose a complex deformation into multiple length-feasible stages.

Angular and Linear Interpolation-based Planner (ALI)

$$\Delta = t - p^0, \quad \sigma = \arg \max_{n=1, \dots, N} \|\Delta_n\|,$$

$$K = \left\lfloor \frac{\|\Delta_\sigma\|}{\lambda} \right\rfloor + 1, \quad \Theta = \theta^t - \theta^p,$$

$$\vartheta = \frac{\Theta}{K}, \quad \delta_\sigma = \frac{\Delta_\sigma}{K}.$$

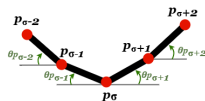
For $\kappa = 1, \dots, K - 1$:

$$p_{\sigma}^{\kappa} = p_{\sigma}^{\kappa-1} + \delta_{\sigma}, \quad \theta_n^{\kappa} = \theta_n^p + \kappa \vartheta_n,$$

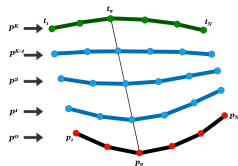
$$\mathbf{p}_n^\kappa = \begin{cases} \mathbf{p}_{n-1}^\kappa + l_s \begin{bmatrix} \cos \theta_n^\kappa \\ \sin \theta_n^\kappa \end{bmatrix}, & n > \sigma, \\ \mathbf{p}_{n+1}^\kappa - l_s \begin{bmatrix} \cos \theta_n^\kappa \\ \sin \theta_n^\kappa \end{bmatrix}, & n < \sigma, \end{cases}$$

Final profile: $p^K = t$.

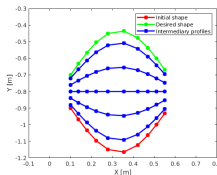
p^0, t : initial and target profiles; σ : maximally displaced point;
 λ : maximum reference step; K : number of stages; l_s : nominal
segment length.



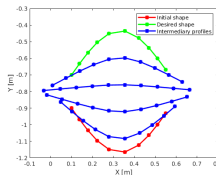
Segment angles



ALI profiles



Pointwise linear interpolation



ALI

Aim: Control the robots' end effectors to follow each intermediary shape.

Constrained shape control

for $\kappa = 1, \dots, K$:

$$\min_{\mathbf{r}} \sum_{n=1}^N \|\mathbf{p}_n^\kappa - \mathbf{p}_n\|,$$

$$\text{s.t.} \quad \dot{\mathbf{p}} = \mathbf{J}\dot{\mathbf{r}}, \quad |\dot{\mathbf{r}}| \leq \boldsymbol{\nu}.$$

$$\mathbf{J} = \text{blkdiag}(\mathbf{J}^1, \mathbf{J}^2), \quad \dot{\mathbf{p}} = \mathbf{J}\dot{\mathbf{r}}.$$

Assignment is determined by nominal cable-order distance, not current Euclidean proximity. $\boldsymbol{\nu}$ contains the componentwise velocity limits.

Diminishing-rigidity Jacobian

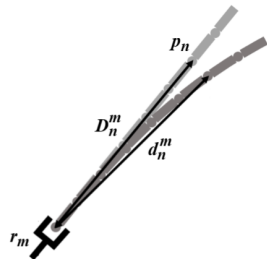
$$D_n^m = \eta_n^m l_s, \quad d_n^m = \left\| \mathbf{p}_n - \begin{bmatrix} r_{mx} \\ r_{my} \end{bmatrix} \right\|,$$

$$\mu_n^m = e^{-|D_n^m - d_n^m|},$$

$$\mathbf{J}_n^m = \mu_n^m \begin{bmatrix} 1 & 0 & -D_n^m \sin \theta_m \\ 0 & 1 & D_n^m \cos \theta_m \end{bmatrix}.$$

m : end effector with pose

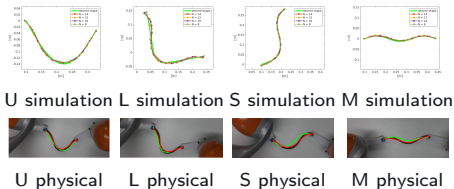
$\mathbf{r}_m = [r_{mx}, r_{my}, \theta_m]^\top$; n : cable point; η_n^m : number of nominal segments from endpoint m to point n ; D_n^m : fully stretched distance; d_n^m : current distance; μ_n^m : slack-dependent geometric weight, not a constitutive material law.



Taut section: $d_n^m \approx D_n^m$, strong coupling.

Slack section: $d_n^m < D_n^m$, diminishing coupling.

Result. The controller uses observed geometry and robot poses; it does not require identified cable stiffness or damping.



Mean pointwise error [mm], $N = 10$	U	L	S	M
Simulation	4.0	6.9	5.2	4.9
Physical experiment	11.7	10.9	10.6	5.2

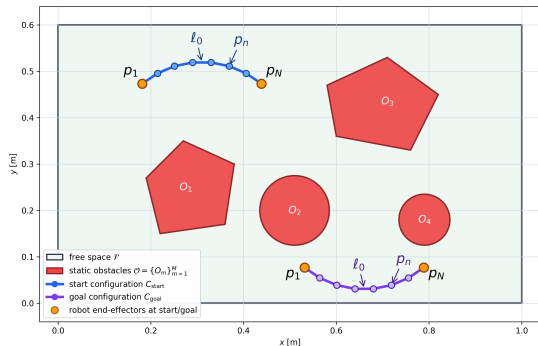
Simulation study

- two cable models, four targets, and $N \in \{8, 10, 12, 14\}$: 32 tasks;
- all tasks completed; six reached the iteration cap before the strict $e_{\text{mean}} < 0.01L$ condition (L : total cable length);
- λ tested from 1 to 100 mm; recommended range: 10–50 mm.

Physical study

- two KUKA LBR iiwa robots and top-view RealSense D435;
- all U/L/S/M tasks completed;
- four tested opposite-concavity tasks completed, with mean errors 4.5–12.4 mm.

Scope: Detailed simulation plots, the DLO model, hardware setup, and full physical error tables are provided in the appendix.



Source: Dissertation, Chapter 4: ordered configuration-space problem.

Configuration:

$$C = (p_1, \dots, p_N) \in \mathbb{R}^{2N}.$$

Find a route

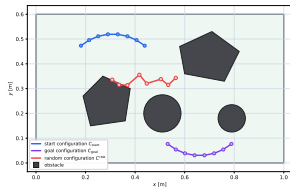
$$\mathcal{P} = (C_0, \dots, C_K), \quad C_0 = C_s, \quad C_K = C_g.$$

Static admissibility

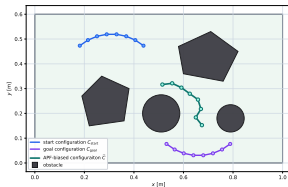
- all points lie inside the workspace;
- complete cable segments avoid obstacles;
- segment lengths remain within bounds.

Transition admissibility

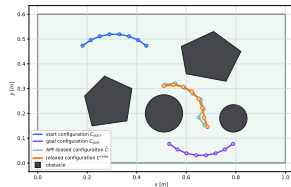
- swept cable-segment regions avoid obstacles;
- point and endpoint ordering is preserved;
- optional robot endpoint constraints are satisfied.



Guided configuration sample



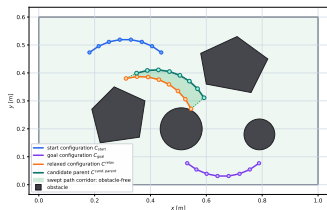
APF-biased candidate



Projected elastic relaxation

1. Grow two search trees from start and goal.
2. Sample complete ordered cable configurations using a goal/tree/broad-random mixture.
3. Use an Artificial Potential Field (APF) to bias each point toward the opposite tree and away from obstacles.
4. Relax the candidate while enforcing workspace, segment-length, and clearance constraints.
5. Accept only static and ordered-transition-valid connections.
6. Join the trees and shortcut the route using the same test.

Result. APF is a sampling bias; the bidirectional tree search performs the global exploration.

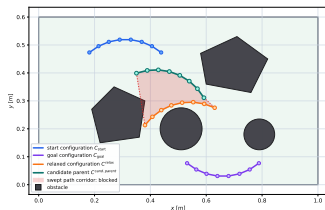


Accepted transition

For segment k , the swept region between configurations C^a and C^b is approximated by

$$\mathcal{S}_k(C^a, C^b) = \text{conv}\{p_k^a, p_{k+1}^a, p_{k+1}^b, p_k^b\}.$$

The transition is rejected if $\mathcal{S}_k(C^a, C^b) \cap \mathcal{O} \neq \emptyset$ for any k .

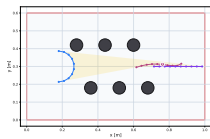


Rejected transition

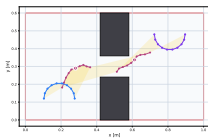
The ordered test also checks:

- rotation of the endpoint axis, preventing the assignment of the two cable ends to the robots from flipping;
- optional end-effector reach and motion bounds.

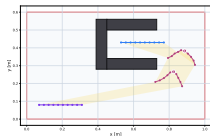
Result. Each neighboring deformation is checked using this swept-polygon approximation, rather than only its two static configurations.



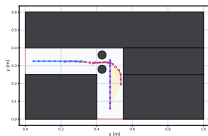
Arc to straight



Narrow gate



U-shaped trap

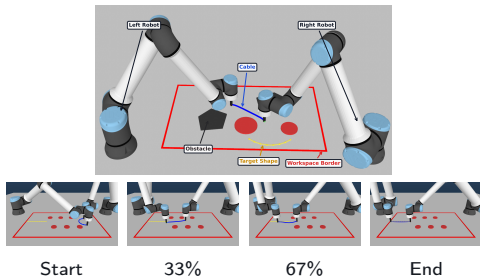


L-corridor

Protocol	10 scenarios, 1000 independent seeds each
Valid routes	10,000 of 10,000 tested runs
Mean time	0.008 ± 0.003 to 1.306 ± 0.871 s
Raw path	13 ± 2 to 25 ± 4 configurations
Refined path	3 ± 1 to 9 ± 1 configurations

Single-threaded C++ implementation.

Result. All tested paths satisfied the stated static and transition predicates. This is empirical evidence, not a completeness guarantee.



Two UR10 manipulators; cable length 0.27 m.

The two robots track the endpoint waypoints of the refined GCR route.

Ten scenarios

- final cable-shape root-mean-square error (RMSE): 1.60–4.66 mm;
- final maximum point error: 2.31–7.81 mm.

Validation boundary: MuJoCo simulation; no physical GCR experiment is claimed.

Main results

1. VFPs reconstruct an ordered point state of a marker-free planar cable from RGB images.
2. ALI generates length-feasible intermediary profiles and enabled four tested opposite-concavity physical transformations.
3. The diminishing-rigidity Jacobian and constrained control use cable geometry without identified stiffness or damping.
4. GCR plans ordered cable routes and rejects transitions whose swept-region approximations intersect obstacles.
5. All 10,000 tested GCR runs met the stated predicates; MuJoCo RMSE was 1.60–4.66 mm.

Established limitations

- planar, quasi-static manipulation;
- no out-of-plane bending or torsion;
- nonlinear elasticity, plasticity, and frictional contact are not modeled;
- clear top view, without occlusion or self-intersection;
- known static obstacles;
- physical validation only for shape control.

Further work

3D perception and manipulation, richer mechanics and contact, and physical global-routing experiments.

Publications

1. K. Almaghout, A. Cherubini, A. Klimchik, *Robotic Co-manipulation of Deformable Linear Objects for Large Deformation Tasks*, Robotics and Autonomous Systems, 2024, 104652.
2. K. Almaghout, A. Klimchik, *Manipulation Planning for Cable Shape Control*, Robotics, 13(1), 2024, 18.
3. K. Almaghout, A. Klimchik, *Deformable Linear Objects Modeling and Manipulation: An Energy-Based Approach*, Advances in Automation V, 2024, pp. 183–194.
4. K. Almaghout, A. Klimchik, *Planar Shape Control of Deformable Linear Objects*, IFAC-PapersOnLine, 55(10), 2022, pp. 2469–2474.
5. K. Almaghout, A. S. Klimchik, *Vision-Based Robotic Comanipulation for Deforming Cables*, Russian Journal of Nonlinear Dynamics, 18(5), 2022, pp. 843–858.

Four works are indexed by Scopus; one is published in a journal included in the list of the Russian Higher Attestation Commission.

Conferences and approbation

1. 10th IFAC Conference on Manufacturing Modelling, Management and Control (MIM), Nantes, France, 2022.
2. International Conference *Nonlinearity, Information and Robotics* (NIR), Innopolis, Russia, 2022.
3. International Russian Automation Conference (RusAutoCon), Sochi, Russia, 2023.

Project support

Russian Science Foundation project No. 22-41-02006.

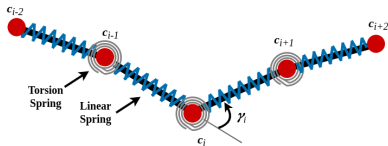
Thank you for your attention

Questions

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Abbreviation	Meaning
DLO	Deformable Linear Object
RGB	Red–Green–Blue image
VFPs	Virtual Feature Points
LI	Linear Interpolation-based Planner
ILI	Improved Linear Interpolation-based Planner
ALI	Angular and Linear Interpolation-based Planner
GCR	Global Cable Routing
APF	Artificial Potential Field
RMSE	Root-mean-square error



- N point masses;
- $N - 1$ stiff linear springs for length preservation;
- $N - 2$ torsion springs for elastic bending;
- damping between adjacent points.

Scope: This is the Chapter 3 simulation environment, not a separate novelty claim and not a source of stiffness or damping parameters for the controller.

Complete force balance:

$$\ddot{\mathbf{p}} = \mathbf{M}^{-1} [\mathbf{f}_l(\mathbf{p}) + \mathbf{f}_b(\mathbf{p}) + \mathbf{f}_d(\mathbf{p}, \dot{\mathbf{p}})].$$

$$E_l = \sum_{n=2}^N \frac{k_l}{2} (\|\mathbf{p}_n - \mathbf{p}_{n-1}\| - l_s)^2, \quad \mathbf{f}_l = -\nabla_{\mathbf{p}} E_l,$$

$$E_b = \sum_{n=2}^{N-1} \frac{k_b}{2} \gamma_n^2, \quad \mathbf{f}_b = -\nabla_{\mathbf{p}} E_b.$$

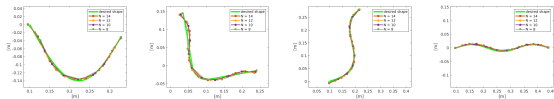
$\mathbf{f}_l$ preserves segment length; $\mathbf{f}_b$ models bending; $\mathbf{f}_d$ suppresses relative oscillation. Backward Euler integration, linearization, and a conjugate-gradient solve are used.

Cable models

- DLO 1: $L = 350$ mm,
 $d = 4$ mm, $E = 100$ MPa;
- DLO 2: $L = 700$ mm,
 $d = 7$ mm, $E = 126$ MPa.

Protocol

- targets: U, L, S, M;
- $N = 8, 10, 12, 14$;
- 32 tasks total;
- strict condition:
 $e_{\text{mean}} < 0.01L$;
- cap: 20K controller iterations.

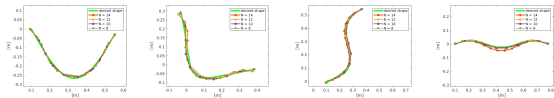


U

L

S

M

DLO 1, all tested N 

U

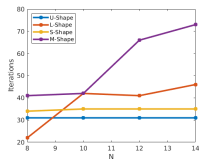
L

S

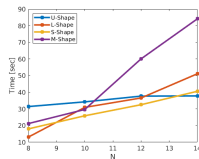
M

DLO 2, all tested N

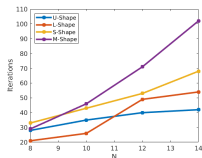
All 32 tasks completed; six reached the iteration cap before satisfying the strict convergence condition.



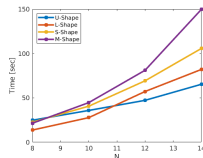
DLO 1: iterations



DLO 1: computation time

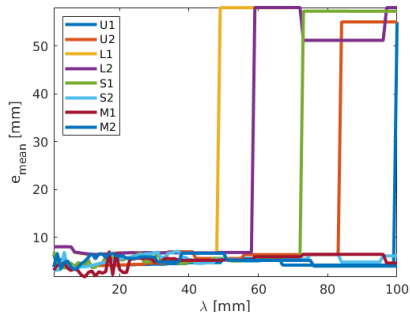


DLO 2: iterations

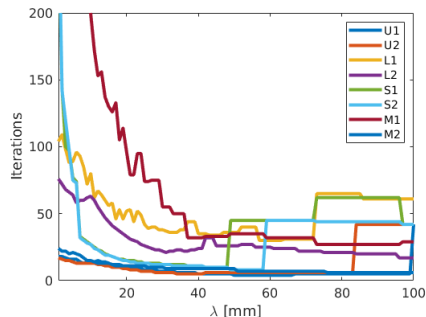


DLO 2: computation time

Result. Iterations and computation time generally increase with N . The six iteration-cap cases occurred for $N = 10, 12, 14$, not for $N = 8$.

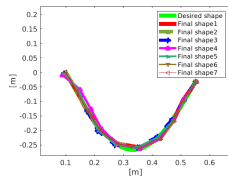


Final mean shape error

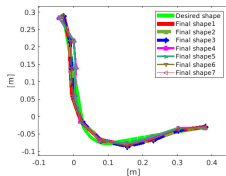


Required iterations

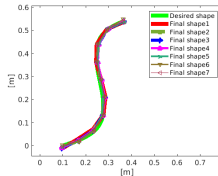
Result. The reference-point step λ was tested from 1 to 100 mm; the reported effective operating range was 10–50 mm.



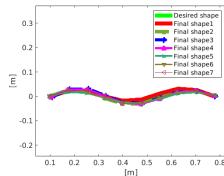
U



L

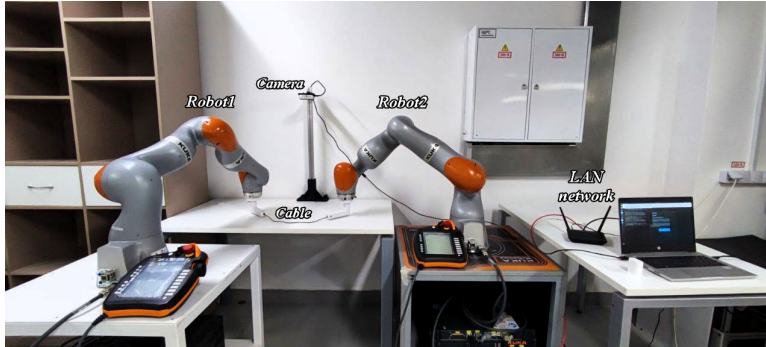


S



M

Target	U	L	S	M
Mean across seven initial shapes [mm]	4.4	7.1	4.3	3.6
Sample standard deviation [mm]	0.5	0.8	1.1	1.2



Source: Dissertation, Chapter 3: physical experimental setup.

- two KUKA LBR iiwa robots;
- top-view Intel RealSense D435;
- cable: $L = 350$ mm, $d = 9$ mm;
- $N = 10$ VFPs;
- ALI step $\lambda = 10$ mm;
- planar, obstacle-free workspace.

Scope: Physical validation of shape control only.



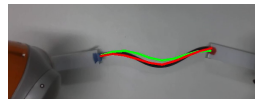
U



L



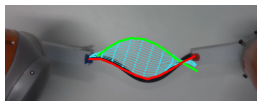
S



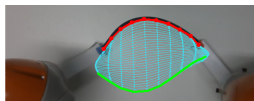
M

		U	L	S	M
Simulation	e_{mean} [mm]	4.0	6.9	5.2	4.9
	e_{std} [mm]	2.0	4.2	2.4	3.0
Physical experiment	e_{mean} [mm]	11.7	10.9	10.6	5.2
	e_{std} [mm]	5.7	5.0	6.4	3.0

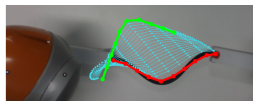
Result. All four physical transformations were completed; measured mean pointwise errors were 5.2–11.7 mm.



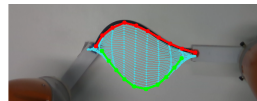
Scenario 1



Scenario 2



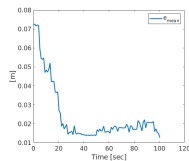
Scenario 3



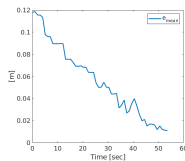
Scenario 4

	Scenario 1	Scenario 2	Scenario 3	Scenario 4
e_{mean} [mm]	12.4	9.0	8.1	4.5
e_{std} [mm]	5.1	3.7	3.1	3.3

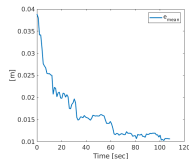
Result. All four tested physical transformations reversed concavity through the planned intermediary profiles.



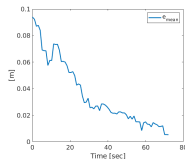
U target



L target



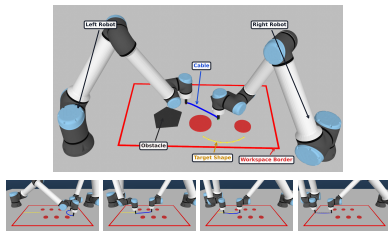
S target



M target

Scenario	Success	Time [s]	Fastest [s]	Iterations [10^3]	Raw	Refined
1	100%	0.240 ± 0.143	0.010	2.08 ± 1.20	17 ± 4	5 ± 1
2	100%	0.283 ± 0.197	0.018	1.75 ± 1.21	20 ± 4	5 ± 1
3	100%	0.008 ± 0.003	0.002	0.07 ± 0.03	13 ± 2	3 ± 1
4	100%	1.306 ± 0.871	0.124	6.98 ± 4.52	22 ± 5	5 ± 1
5	100%	0.017 ± 0.005	0.005	0.09 ± 0.03	16 ± 3	7 ± 1
6	100%	0.145 ± 0.061	0.014	0.95 ± 0.42	18 ± 4	7 ± 1
7	100%	0.151 ± 0.066	0.020	1.36 ± 0.57	19 ± 4	7 ± 1
8	100%	0.072 ± 0.033	0.008	0.65 ± 0.31	16 ± 3	6 ± 1
9	100%	0.104 ± 0.059	0.004	0.72 ± 0.41	18 ± 3	5 ± 1
10	100%	0.339 ± 0.139	0.054	2.60 ± 1.08	25 ± 4	9 ± 1

Each row summarizes 1000 independent seeds. Success denotes paths satisfying the stated static and transition predicates.



Scenario	RMSE [mm]	Maximum [mm]
1	2.20	3.41
2	2.22	3.72
3	4.32	6.49
4	1.60	2.31
5	1.87	2.70
6	2.08	3.32
7	4.66	7.81
8	4.13	6.28
9	4.58	7.32
10	2.83	4.73

Scope: Two UR10 manipulators, a 0.27 m cable; MuJoCo simulation only.

Required component	Content of the dissertation
Mathematical modeling	Ordered DLO state; mass–spring simulation model; diminishing-rigidity Jacobian; deformation-path model; ordered configuration-space model.
Numerical methods	VFPs; LI, ILI, and ALI; constrained shape control; bidirectional GCR; APF bias; projected relaxation; swept-transition tests.
Software packages	Perception and control software; MATLAB simulation studies; single-threaded C++ GCR implementation; MuJoCo validation.

Related specialty passport points: 2 (effective computational methods), 3 (problem-oriented programs), 5 (validation methods), 6 (simulation systems), 8 (comprehensive investigation), and 9 (numerical experiments).

Obj.	Developed result	Verification	Passport points
1	Ordered DLO state and VFP perception	Physical KUKA image-feedback experiments	2, 3, 5
2	LI/ILI/ALI; ALI discrete segment-length invariant	MATLAB studies and four physical concavity reversals	2, 5, 8
3	Diminishing-rigidity Jacobian and constrained controller	Simulation and hardware shape errors	2, 5, 8
4	GCR model, search, and ordered swept-transition test	10,000 benchmark runs and MuJoCo	2, 6, 8
5	MATLAB, C++, perception, and robot-control research software	Parameter studies, statistics, hardware, and simulation	3, 6, 9

Comparison basis. Selected Chapter 3 works addressing comparable DLO shape-control tasks; the table is not a universal ranking.

Approach	Space	Robot	Cable marks	Demonstrated task scope
Zhu et al. (2018)	2D	dual-arm	no	local deformation
Zhu et al. (2021)	2D	single-arm	no	local deformation
Qi et al. (2021)	2D	dual-arm	no	local deformation
Qi et al. (2022)	2D	dual-arm	no	local deformation
Wang et al. (2022)	2D	dual-arm	no	local deformation
Yu et al., Shape control (2022)	2D/3D	single-arm	yes	large deformation
Yu et al., Global model (2022)	2D/3D	dual-arm	yes	large deformation
Proposed approach	2D	dual-arm	no	large deformation and concavity change

Within the selected works, the proposed method combines marker-free dual-arm control, large planar deformation, and opposite-concavity tasks.

Approach	Intermediate profiles	Robot–DLO relation	Robot-motion method
Zhu et al. (2018)	none	numerical Jacobian	proportional control
Zhu et al. (2021)	linear interpolation	numerical Jacobian	proportional control
Qi et al. (2021)	none	numerical Jacobian	finite-time sliding-mode control
Qi et al. (2022)	none	numerical Jacobian	model-predictive control
Wang et al. (2022)	none	hybrid offline–online model	model-predictive control
Yu et al., Shape control (2022)	none	hybrid offline–online model	adaptive control
Yu et al., Global model (2022)	none	hybrid offline–online model	optimization-based adaptive control
Proposed approach	angular–linear interpolation	diminishing-rigidity Jacobian	constrained optimization

The proposed Jacobian is an analytical geometric approximation of slack-dependent coupling; it is not an identified elastic constitutive law.